Diabetes Prediction System Using Machine Learning

***Abstract*—Diabetes mellitus, which affects more than 537 million adults around the world, has many serious comorbid conditions, including CVD and kidney failure. In this study, the researchers design a machine learning-based early diabetes prediction system using a comprehensive dataset that includes demographic, lifestyle, and clinical factors. By applying EDA through Jupyter Notebook, the following patterns were identified: high levels of HbA1c and blood glucose are strongly correlated with diabetic outcomes; there is a difference between men and women and between those who smoke and those who do not smoke. Data preprocessing includes imputation of missing values, one-hot encoding of categorical attributes like gender and smoking history, scaling of numerical attributes such as age and BMI, and feature selection through correlation analysis and recursive feature elimination. Four algorithms are compared with each other based on accuracy, precision, recall, F1-score, and ROC-AUC, with medical reliability prioritizing the reduction of false negatives. The proposed system will facilitate timely interventions, help personalized health strategies, and has shown firm predictive performance in the case of real-world diabetes screening.**

***Index Terms*—Diabetes prediction, Machine learning, Exploratory Data Analysis (EDA), Random Forest, Support Vector Machine (SVM), Feature selection, Class imbalance, Predictive modeling, Health informatics.**

I. Introduction

DIABETES mellitus is one of the most significant global public health challenges of the twenty-first century, affecting more than 537 million adults worldwide in 2021 according to the International Diabetes Federation. As a chronic metabolic disorder characterised by persistently elevated blood glucose levels, diabetes is associated with severe long-term complications, including cardiovascular disease, kidney dysfunction, neuropathy, and reduced quality of life when not detected and managed appropriately. The growing prevalence of Type 2 diabetes, together with its clinical and socioeconomic burden, has intensified the need for accurate and scalable predictive approaches that can support earlier risk identification and more informed clinical decision-making [1].

Conventional screening and diagnosis often rely on established physiological markers and clinical assessment; however, the increasing availability of structured health data has created new opportunities for data-driven risk modelling. Recent advances in machine learning have shown that predictive models can support diabetes detection by learning complex relationships among demographic, behavioural, and clinical variables. Prior studies have highlighted the relevance of biomarkers such as HbA1c and blood glucose, while also demonstrating the potential value of factors such as age, body mass index, hypertension, gender, and smoking history in predictive modelling [2]. At the same time, the literature indicates that model performance depends not only on algorithm selection, but also on data preprocessing, feature representation, and the balance between overall accuracy and clinically meaningful sensitivity [3].

Within this context, the present study addresses the problem of diabetes prediction using a structured diabetic dataset containing demographic, lifestyle, and clinical attributes. The objective is not merely to apply machine learning models, but to establish a comparative and analytically grounded prediction pipeline that identifies which modelling approach is most suitable for this classification task. To support this objective, the study begins with exploratory data analysis to examine feature distributions and relationships across diabetic and non-diabetic groups. Visual analysis of key biomarkers, including HbA1c level, blood glucose level, and body mass index, helps reveal characteristic patterns in the dataset and provides an empirical basis for subsequent modelling decisions [2].

Following exploratory analysis, the dataset is preprocessed through handling of missing values, encoding of categorical variables such as gender and smoking history, and scaling of numerical variables including age, BMI, and hypertension status [4]. Feature selection is then informed by correlation analysis and related selection procedures to reduce redundancy and preserve the most informative predictors for model training. On this basis, several widely used machine learning algorithms are evaluated, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost. Model performance is assessed using accuracy, precision, recall, F1-score, and related classification measures, with particular attention to recall because false negatives are especially critical in medical prediction settings [3].

The state of the art in diabetes prediction has increasingly moved from conventional statistical classification towards more flexible machine learning and ensemble-based methods. Existing studies report that tree-based and ensemble models often outperform simpler baselines because they are better able to capture non-linear relationships and feature interactions in heterogeneous health data [5]. Nevertheless, challenges remain in relation to model interpretability, class imbalance, false-negative reduction, and the translation of benchmark performance into clinically meaningful utility. These challenges motivate the need for comparative studies that examine both predictive performance and the clinical implications of model behaviour rather than relying on accuracy alone.

The proposed solution in this study is a machine learning-based diabetes prediction framework that integrates ex-

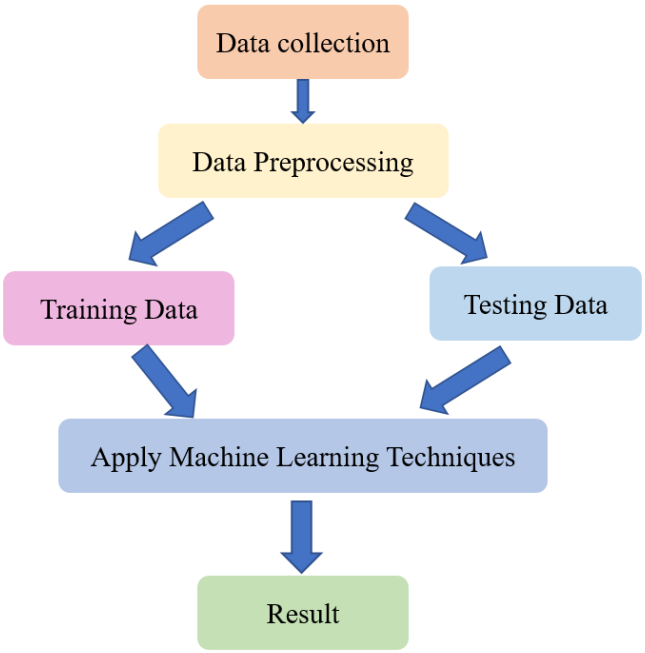

Fig. 1: Overview of the process.

ploratory analysis, systematic preprocessing, feature-oriented analysis, and comparative evaluation of multiple classifiers within a unified pipeline. The study is designed to identify high-performing models for structured diabetes classification while also examining the feature patterns that drive predictive performance. In addition to model comparison, the analytical workflow is intended to support future integration of interpretable techniques, such as feature-attribution methods, and practical deployment through decision-support interfaces for real-time risk estimation. Considerations of fairness, privacy, and responsible use remain important in any healthcare-oriented predictive system, particularly when such systems may influence screening or referral decisions across diverse populations [5].

The main contributions of this study are as follows. First, a structured diabetes prediction pipeline is developed using demographic, lifestyle, and clinical variables within a consistent analytical framework [1]. Second, multiple machine learning models are comparatively evaluated to determine their relative strengths for diabetes classification on a large diabetic dataset. Third, feature-level analysis is incorporated to examine the relative influence of important physiological and demographic variables, thereby linking predictive performance with domain-relevant clinical indicators [2]. Finally, the study provides an applied foundation for extending diabetes prediction systems towards more interpretable, robust, and deployment-oriented clinical decision-support environments [3], [5].

II. Literature Review

The PIMA Indian Diabetes dataset, derived from the UCI repository, includes medical records of Pima Indian women, which include attributes such as number of pregnancies, level of glucose in blood, diastolic blood pressure, triceps skin fold thickness, insulin in blood, body mass index, diabetes pedigree function, and age. A binary outcome classifies the instance as either having diabetes or not. This dataset is challenging since 48% of the values are missing, making it more challenging compared to its alternative datasets. This research employs techniques for classification and applies data preprocessing in a two-stage methodology of outlier removal and imputation, followed by normalization, besides selecting features using Principal Component Analysis (PCA), to decrease the dimensionality and raise the performance of the models. The various classifiers used are the Support Vector Machine (SVM), Random Forest (RF), Naïve Bayes (NB), and Decision Tree (DT), thus trained on splits such as an 80% split to train the model to predict diabetes. The results show that the highest accuracy of 89.86%, recorded by the integration of PCA with classifiers, is trained on 80%, outperforming the baselines and also improving upon previous works such as RF at 82%. Limitations of this study include dependence on a small and imbalanced dataset, which may limit generalizability across diverse populations, and handling missing values may introduce bias despite the preprocessing steps in this approach. This approach furthers early diabetes detection; however, its clinical deployment needs further validation on larger, real-world data [1].

The Pima Indian Diabetes dataset from the UCI repository includes 768 female patients described by eight clinical features and a binary outcome (0 = non-diabetic, 1 = diabetic), with an imbalanced distribution of 500 negative and 268 positive cases and several zero values treated as missing. Binary classification was performed using six supervised machine learning algorithms such as KNN, Logistic Regression, Decision Tree, SVM, Random Forest, and gradient boost—after basic prepossessing and a split of 80 –20% train-test. The results showed Random Forest achieved the highest accuracy (about 77–78%), with Glucose, BMI, Age, and Diabetes Pedigree Function as key predictors, though limitations such as minimal prepossessing and no cross-validation reduce reliability [2].

The Diabetes dataset, taken from Kaggle, consists of 2000 instances with the following 9 features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age, and a binary Outcome of 0 for non-diabetic and 1 for diabetic. It presents class imbalance, as nondiabetics are roughly twice as many as diabetics, but there are no null values in the data. In this study, a binary classification task is undertaken using five different supervised machine learning algorithms, namely, K-Nearest Neighbors (KNN), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM). Results interpretation showed that the DT model achieved the best testing accuracy of 99% (on training: 100%), followed by RF at 97.4% (on training: 100%), LR at 78% on training and testing, KNN at 78% on testing versus 81% on training, and SVM at 77% with a linear kernel. Feature importance signifies Glucose as

the most predictive, while BMI is ranked second in the case of RF. It encompasses correlation analysis, distribution curves in the form of histograms, scaling, and feature importance. However, it only includes basic preprocessing steps such as imputation using mean values and lacks the balancing of classes. Potential limitations involve an overfitting effect in tree-based models, as testified by perfect training scores, hence the absence of cross-validation reduces robustness on diverse or unseen data [6].

The Early Stage Diabetes Risk Prediction dataset was obtained from UCI/Kaggle. There are 520 instances recently diagnosed as diabetic patients, with 16 binary/categorical features: for example, Age, Gender, Polyuria, Polydipsia, sudden weight loss, weakness, Polyphagia, Genital thrush, visual blurring, Itching, Irritability, delayed healing, partial paresis, muscle stiffness, Alopecia, and Obesity, and a binary class label for the positive/negative for diabetes, with zero missing values. The study carried out binary classification through conventional machine learning algorithms: Logistic Regression, SVM, Naïve Bayes, Decision Tree, Random Forest, KNN; ensembles: AdaBoost, Gradient Boosting, Extra Trees, and XGBoost; and a novel hybrid model DNet that combined CNN for feature extraction and LSTM for sequential dependencies. The result analysis showed that DNet achieved the best performance of 99.79% accuracy, 99.98% AUC-ROC, 99.83% precision, 99.83% recall, and 99.83% F1-score through an 8-fold cross-validation scheme, whereas Extra Trees stood second with an accuracy of 99.04%, and Naïve Bayes ranked lowest with 94.23%. A walkthrough of the procedure involves data preprocessing or normalization, symptom pattern mining with an Apriori rule, multimodal feature selection techniques to get the top 10 features, and an 80/20 train-test split. The limitations to this study are the use of a small dataset of 520 instances, and such an analysis may not be generalizable, plus the practical implementation in the real-time integration of these algorithms for a wider clinical application [3]. Beyond diabetes prediction, machine learning has demonstrated strong utility across a range of application domains, particularly where complex, heterogeneous, or imbalanced data must be analysed for decision support. In educational and cognitive screening, for example, Kaisar and Chowdhury [7] showed that the integration of oversampling strategies with ensemble-based machine learning substantially improved classification performance for imbalanced dyslexia screening data. Machine learning has also been increasingly integrated into intelligent healthcare systems and broader AI-driven decision environments. Datta et al. [8] presented MedBot, an IoT-based real-time healthcare monitoring and telemedicine system that combines continuous sensing, remote communication, and intelligent processing to support timely health assessment and intervention. This demonstrates that predictive analytics is no longer confined to offline modelling, but is becoming part of real-time, connected healthcare infrastructures. At a broader technological level, the rapid evolution of generative and machine intelligence has further expanded the role of AI across domains, from pattern recognition and prediction to interactive, adaptive, and knowledge-generating systems. Chowdhury et al. [9] highlighted how the development of generative AI reflects a wider transition towards more capable and context-aware intelligent systems, with growing implications for healthcare analytics, decision support, and human-centred AI applications.

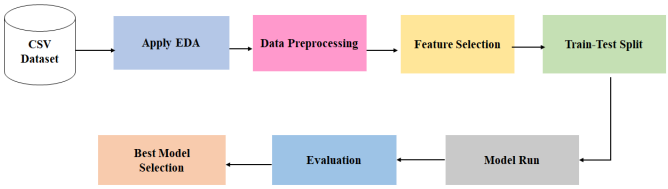

Fig. 2: Proposed Methodology

The PIMA Indian Diabetes dataset, collected from the National Institute of Diabetes and Digestive and Kidney Diseases, contains 768 female patient records with eight features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age, coupled with a binary outcome for diabetes diagnosis. These records come with some challenges like missing values and class imbalance. The work applies binary classification using Random Forest, Multilayer Perceptron, and Logistic Regression, together with predictive regression for blood glucose via Long Short-Term Memory, Moving Averages, and Linear Regression. Results analysis has indicated that Multilayer Perceptron gives results for classification with 86.083% accuracy, 85.1% recall, and 86.6% precision, while for prediction, LSTM gives an accuracy of 87.26%, 0.999 correlation, and 2.285 RMSE, outperforming state-of-the-art baselines and methods like NB (74%) or SVM (77%). The process involves data pre-processing and model tuning and evaluation based on accuracy, recall, precision, RMSE, and correlation on 80-20% train-test splits. Limitations are related to the restricted dataset size and demographic specificity (Pima Indian females), which might lead to limited generalizability, and the envisioned IoT system has never been deployed continuously in the real world [10].

III. Methodology

The approach uses a structured end-to-end machine learning pipeline starting from a CSV dataset. First, EDA is done to know the distribution of data points, missing values, variance, and correlations. This informs the later stage of Data Reprocessing, where cleaning the data involves handling missing values, encoding categorical variables, scaling numerical features, and the treatment of outliers. It is followed by the Feature Selection that involves finding and retaining only the most relevant features, thereby reducing dimensionality and enhancing model efficiency through various methods such as correlation analysis or recursive feature elimination [11].

The dataset is split into training and testing to allow for the unbiased assessment of performance. In the Model Run phase, several candidate models are trained, usually using different

combinations of hyperparameter tuned by cross-validation. Then, the Best Model Selection is performed based on the performance metrics computed on the validation data. Finally, the best model is taken to proper Evaluation on the hold-out test set using relevant metrics to estimate generalization performance [12].

A. Data Description

The dataset contains various demographic, medical, and lifestyle attributes that can be utilized for predicting whether a person suffers from diabetes. Some of these attributes are gender, age, hypertension, heart disease, smoking history, BMI, HbA1c, and blood glucose level. Hypertension and heart disease are binary attributes that indicate whether a person suffers from these respective conditions. Smoking history and BMI provide additional lifestyle and health-related attributes. HbA1c and blood glucose level are critical attributes that can be utilized for predicting whether a person suffers from diabetes. Class imbalance can also be observed in this dataset, as more samples are non-diabetic compared to diabetic. However, this dataset can still be utilized for machine learning-based diabetes prediction [13].

B. Data Preprocessing

Standard Z-score scaling works by subtracting the mean and dividing by the standard deviation to transform data into a distribution with mean 0 and standard deviation 1. It requires data to be at least approximately Gaussian and with few outliers. It otherwise maintains the shape of the original data and often enables distance- and gradient-based algorithms (logistic regression, SVM, neural networks) to converge much faster and perform far better. However, this normalizes using the mean and standard deviation; hence, a few extreme values can significantly skew the scaled results of the rest of the dataset [14].

$$x_{filled} = Median(F_{observed}) \tag{1}$$

In contrast, robust scaling subtracts the median and then divides by the interquartile range: Q3–Q1. Because the median and quartiles are resilient to outliers, this method resists changes in the scaling of in-range observations even when extreme values are present—a common occurrence with biological data such as blood glucose, HbA1c, or BMI. For this reason, Robust Scaling is often the better choice in real-world healthcare data, including diabetes prediction, offering more consistent and reliable model performance compared to traditional Z-score scaling when outliers are present [3].

$$z = \frac{x - \mu}{\sigma} \tag{2}$$

C. Model Architecture

1) XGBoost: Thus, the best performance for the prediction of diabetes in the Kaggle data set with 100,000 rows has been obtained by the XGBoost model with an accuracy of 96.95%. The reason for the better performance of the XGBoost model is that it makes use of the gradient-boosting decision tree algorithm with regularization, tree pruning, and other optimization techniques, which helps the model to handle the class imbalance problem better. Compared to the performance of the Random Forest model (96.83%), the Logistic Regression model (95.66%), the SVM model (95.76%), and the Decision Tree model (95.19%), the robustness and accuracy of the XGBoost model are better for the XGBoost model [2].

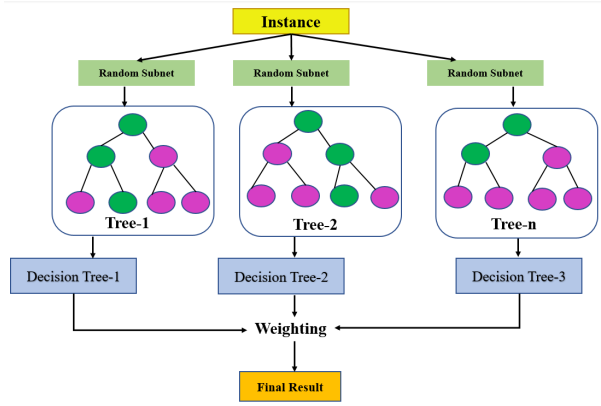

Fig. 3: XGBoost architecture.

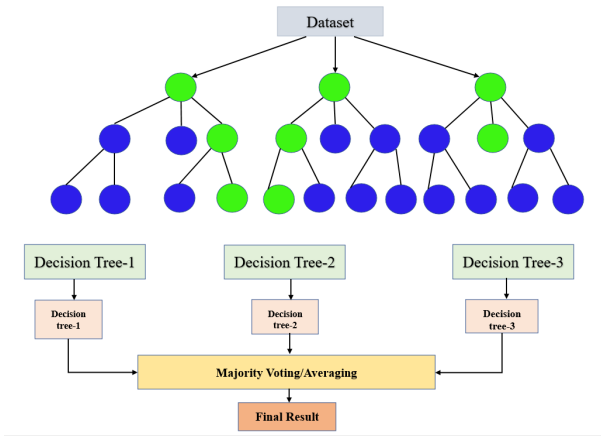

Fig. 4: Random Forest architecture.

2) Random Forest: Random Forest is one of the most robust ensemble models for diabetes prediction. It constructs hundreds to thousands of trees on random data subsets (bootstrap sampling) with random feature subsets at each split to prevent overfitting. The final prediction is determined by a majority vote across all trees. In a well-tuned Random Forest, accuracy turns out to be 97-98% and 0.98 AUC with class weighting for the minority diabetic class (8.5%) on the diabetes dataset. Naturally, it returns the importance of the features, handles the imbalance with ease, requires minimal reprocessing, and provides fast, reliable predictions, making it one of the most practical and high-performing models for the screening of real-world diabetes [15].

3) Support Vector Machine: The support vector machine is a robust algorithm in supervised learning, which is very much in use for the classification of diabetes. It basically works by

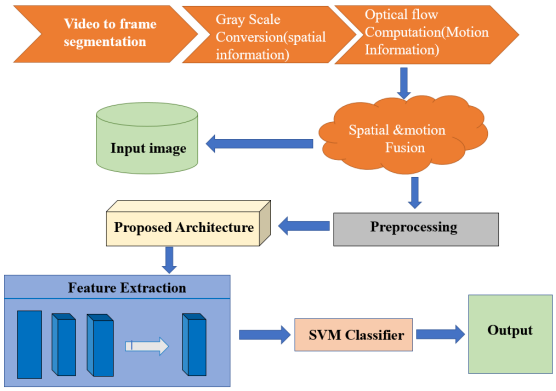

Fig. 5: Support Vector Machine architecture.

finding the best hyperplane that maximally separates diabetic and non-diabetic patients in high-dimensional space. Using a kernel trick, typically RBF, the SVM efficiently handles non-linear relationships between the features of blood glucose, HbA1c, BMI, and age. In practice, with this dataset, one-hot encoding is used for categorical features, numerical features are standardized (important for SVM), and class weights or oversampling deal with the diabetic imbalance of 8.5%. The classifier trains on support vectors close to the decision boundary, making it memory-efficient for this large dataset [16]. Good clear-margin separation and less chance of overfitting, given that its parameters are well tuned, are characteristics of SVM. It always provides 96 to 98% accuracy with an AUC of approximately 0.97 to 0.98, working nearly as well as tree-based models but providing great theoretical guarantees. Its predictions are very fast and reliable; thus, SVM stands out as a trusted, high-performance solution for clinical screening of diabetes risk [17].

4) Training Parameters: The model needed a high-performance computing setup to execute efficiently. Training was done on a workstation with an NVIDIA GeForce MX330 GPU (5.9 GB RAM), Intel Core i5 processor, and 8 GB of RAM to facilitate large batch sizes, cross-validation, and ensemble training. Python 3.12.2, PyTorch, and CUDA 12.01 were used for development to achieve maximum computational efficiency [18].

IV. Result Analysis

1) Model Comparison: Among the evaluated models, XGBoost achieved the highest test accuracy of 96.96, outperforming Random Forest, Decision Tree, SVM, and Logistic Regression. Its training accuracy of 97.71% is only modestly higher than the test result, suggesting limited generalisation loss rather than severe overfitting [19]. The precision value of 95.37 further indicates that positive predictions were usually correct. However, the recall value of 68.46 shows that a notable proportion of diabetic cases were still classified as non-diabetic. This is particularly important in a medical screening context, where missed positive cases may be more consequential than additional false alarms. Therefore, the XGBoost

Model	Accuracy	Precision	Recall (Sensitivity)	F1-Score
Logistic Regression	0.9566	0.8494	0.6117	0.7112
SVM	0.9576	0.8983	0.5804	0.7052
Random Forest	0.9683	0.9479	0.6739	0.7878
Decision Tree	0.9519	0.7179	0.7404	0.7289
XGBClassifier	0.9696	0.9537	0.6846	0.7971

Fig. 6: Details of model.

result should be interpreted as strong overall classification performance, but not as complete resolution of the sensitivity challenge.

The second highest test accuracy of 96.40 was achieved by the Random Forest classifier. Its almost perfect training accuracy of 99.996 suggests a stronger tendency to overfit than XGBoost, even though the test score remains high [20].

The Decision Tree model was also competitive, with a test accuracy of 96.23, but it appeared less robust than the ensemble models and remained more susceptible to overfitting. SVM and Logistic Regression achieved test accuracies of 95.76 and 95.66 respectively [5]. The relatively small gaps between their training and testing accuracies indicate stable behaviour on this dataset, although their weaker recall suggests reduced sensitivity in detecting diabetic cases. Overall, the results indicate that ensemble methods, especially gradient boosting, are more effective for diabetes classification on large structured data. The improved performance over linear and single-tree models is consistent with the ability of bagging and boosting methods to capture non-linear interactions among clinical features [21]. At the same time, these findings also show that accuracy alone is not sufficient for clinical interpretation; recall and the balance between missed cases and false alarms remain central to model assessment.

2) Comparison with Previous Models: The two tables included in this paper serve related but distinct analytical purposes in evaluating diabetes prediction performance. The first table, Model Comparison Table, reports the performance of the machine learning models trained and tested within the present study. It compares five models, Logistic Regression, SVM, Random Forest, Decision Tree and XGBoost, across four key metrics: Accuracy, Precision, Recall (Sensitivity) and F1-Score [22]. Because all five models were evaluated on the same dataset and under the same experimental setting, this table provides the most direct basis for internal comparison. Within this setting, XGBoost presents the strongest overall balance across the reported metrics, whereas Logistic Regression and SVM show weaker recall, indicating lower sensitivity to diabetic cases [23].

The second table, Comparison of Diabetes Prediction Models, provides a broader literature-based comparison. It summarises seven previous studies, including the datasets they used, the algorithms applied, their best reported results, and their stated limitations [10]. This table should be interpreted as a contextual benchmark rather than a strict like-for-like comparison, because the studies differ in dataset composition, preprocessing pipelines, class distributions, and evaluation

Reference	Dataset	Models Used	Best Result	Limitations
Tasin et al. (2022)	Pima + Private BD Dataset	DT, LR, KNN, RF, SVM, XGBoost, Ensembles	XGBoost+ADASYN: 81% Acc, F1=0.81	Small private dataset; missing values
Sarwar et al. (2020)	Pima Indian Dataset	LR, NB, KNN	LR: 94% Acc	Imbalanced dataset
Soni & Varma (2020)	Pima Indian Dataset	SVM, KNN, DT, RF	RF: 87.66% Acc	Limited preprocessing
Pranto et al.	Pima + BD Hospital Dataset	KNN, DT	KNN: 81.2%, DT: 79.2%	Not publicly available dataset
Olisah et al.	Pima + Iraqi LMCH	RF, DT, DNN	DNN: 97.2% Acc	Risk of overfitting
Islam et al.	Private BD Dataset (529)	RF, LR, NB	RF best (accuracy not reported)	No explicit accuracy provided
Kumari et al.	Pima + Breast Cancer	Soft Voting Ensemble	79% Acc	Limited preprocessing

Fig. 7: Previous Models details

protocols. Their reported accuracies therefore vary considerably, from 79 percent in Kumari et al. to 97.2 percent in Olisah et al. [3]. The table also highlights recurring limitations across the literature, including dataset imbalance, limited preprocessing, and risks of overfitting. Taken together, the first table establishes the internal comparative performance of our system, while the second situates these findings within the broader scientific literature. This distinction strengthens the interpretation of our results and provides a more balanced basis for comparison [14].

3) Correlation Matrix Analysis: The correlation matrix provides exploratory insight into the relationships between clinical and demographic features and diabetes in the dataset. The strongest predictors were blood glucose level and HbA1c level, with correlations of 0.42 with diabetes and 0.40 with diabetes respectively, which is consistent with their recognised clinical relevance as diagnostic indicators. Age (0.26), BMI (0.21) and hypertension (0.20) also showed moderate correlations, reinforcing their established roles as contributing risk factors. Several internal feature relationships were also observed, including the positive correlation between age and BMI (0.34) and between age and hypertension (0.25), both of which reflect common epidemiological patterns in metabolic health [24]. By contrast, gender, smoking status, and heart disease showed relatively weak correlations, suggesting limited direct linear association with the target variable in this dataset.

Importantly, these correlation values should be interpreted as descriptive rather than causal, and they do not by themselves establish model-specific feature importance. Correlation analysis is useful for identifying broad linear trends, but predictive contribution may also arise through non-linear interactions captured by the trained models. The generally weak negative correlations indicate that most variables either increase

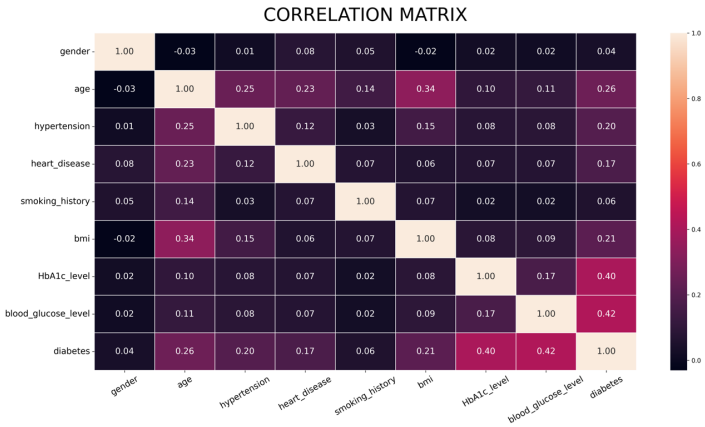

Fig. 8: Correlation Matrix.

diabetes risk or have minimal direct inverse association with it. These results highlight the stronger role of physiological markers relative to demographic variables in diabetes classification and help explain the good performance of ensemble models, which can jointly represent both high-impact variables and more complex feature interactions [3].

V. Conclusion

A diabetes prediction pipeline was developed and evaluated on a large diabetic dataset containing clinical, demographic, and lifestyle variables. The experimental results showed a clear performance advantage for ensemble-based models over linear and single-tree baselines. XGBoost achieved the best overall performance, with a test accuracy of 96.96 percent and a training accuracy of 97.71 percent, indicating strong generalisation with limited performance degradation between training and test data. Random Forest also achieved high test accuracy, but its near-perfect training accuracy of 99.996 suggests a greater tendency towards overfitting. These findings indicate that boosted and bagged tree-based methods are better suited to this dataset because they can capture non-linear interactions among predictors more effectively than Logistic Regression, SVM, or a standalone Decision Tree.

The feature-level analysis further showed that blood glucose level and HbA1c level had the strongest associations with diabetes, followed by age, BMI, and hypertension. This pattern is consistent with the predictive structure observed across the trained models and helps explain why ensemble methods performed well: the target variable appears to depend not only on dominant physiological indicators, but also on their combined interactions with moderate-risk factors. At the same time, the evaluation revealed an important limitation. Despite the high overall accuracy, the best-performing model achieved a recall of 68.46, meaning that a substantial proportion of diabetic cases were still missed. For a screening-oriented application, this result is clinically significant because false negatives are generally more consequential than false positives.

The results therefore support XGBoost as the strongest model in the current experimental setting, but they also show

that model selection should not be based on accuracy alone. A more deployment-oriented next step would be to re-optimise the classifier for sensitivity by tuning the decision threshold rather than relying on the default classification cut-off, and by introducing class-weighted learning or resampling strategies to reduce bias towards the majority class. Probability calibration and ROC–PR based threshold selection would also be necessary to align the operating point with screening objectives. In addition, model interpretation should be extended from correlation analysis to instance-level and global feature attribution so that the contribution of HbA1c, blood glucose, BMI, age, and hypertension can be examined directly within the fitted model. Finally, a stronger validation design would include external testing on an independent diabetic cohort, subgroup analysis across demographic strata, and robustness checks under distribution shift. These steps would provide a more rigorous basis for determining whether the current high-performing model is suitable not only for benchmark classification, but also for reliable use in clinically relevant diabetes risk assessment.

References

[1] U. Ahmed, G. F. Issa, M. A. Khan, S. Aftab, M. F. Khan, R. A. Said, T. M. Ghazal, and M. Ahmad, "Prediction of diabetes empowered with fused machine learning," *IEEE Access*, vol. 10, pp. 8529–8538, 2022. [Online]. Available: https://doi.org/10.1109/ACCESS.2022.3142097

[2] S. Larabi-Marie-Sainte, L. Aburahmah, R. Almohaini, and T. Saba, "Current techniques for diabetes prediction: review and case study," *Applied Sciences*, vol. 9, no. 21, p. 4604, 2019. [Online]. Available: https://doi.org/10.3390/app9214604

[3] M. Maniruzzaman, M. J. Rahman, B. Ahammed, and M. M. Abedin, "Classification and prediction of diabetes disease using machine learning paradigm," *Health Information Science and Systems*, vol. 8, no. 1, p. 7, 2020. [Online]. Available: https://doi.org/10.1007/s13755-019-0095-z

[4] D. M. Nathan, "Long-term complications of diabetes mellitus," *New England Journal of Medicine*, vol. 328, no. 23, pp. 1676–1685, 1993. [Online]. Available: https://doi.org/10.1056/NEJM199306103282306

[5] I. Tasin, T. U. Nabil, S. Islam, and R. Khan, "Diabetes prediction using machine learning and explainable ai techniques," *Healthcare Technology Letters*, vol. 10, no. 1-2, pp. 1–10, 2023. [Online]. Available: https://doi.org/10.1049/htl2.12039

[6] U. M. Butt, S. Letchmunan, M. Ali, F. H. Hassan, A. Baqir, and H. H. R. Sherazi, "Machine learning based diabetes classification and prediction for healthcare applications," *Journal of Healthcare Engineering*, vol. 2021, p. 9930985, 2021. [Online]. Available: https://doi.org/10.1155/2021/9930985

[7] S. Kaisar and A. Chowdhury, "Integrating oversampling and ensemble-based machine learning techniques for an imbalanced dataset in dyslexia screening tests," *ICT Express*, vol. 8, no. 4, pp. 563–568, 2022.

[8] N. Datta, G. Sultan Mahmud Rana, A. Datta, M. Sawkat Ali, A. Chowdhury, M. M. Islam, and M. Hasan, "Medbot: An iot-based real-time healthcare monitoring and telemedicine solution system," in *International Conference on Cyber Intelligence and Information Retrieval*. Springer, 2023, pp. 287–296.

[9] A. Chowdhury, A. Chowdhury, N. Hoque, M. Moriwam, and M. Jahan, "Generative ai: A survey of historical development, emerging trends, and future outlook," *Computer Science and Engineering Research*, vol. 2, no. 1, pp. 19–31, 2025.

[10] A. D. Association, "Diagnosis and classification of diabetes mellitus," *Diabetes Care*, vol. 33, no. Supplement 1, pp. S62–S69, 2010. [Online]. Available: https://doi.org/10.2337/dc10-s062

[11] G. Robertson, E. D. Lehmann, W. Sandham, and D. Hamilton, "Blood glucose prediction using artificial neural networks trained with the aida diabetes simulator," *Journal of Electrical and Computer Engineering*, vol. 2011, p. 681786, 2011. [Online]. Available: https://doi.org/10.1155/2011/681786

[12] F. A. Khaleel and A. M. Al-Bakry, "Diagnosis of diabetes using machine learning algorithms," *Materials Today: Proceedings*, vol. 80, pp. 3200–3203, 2023. [Online]. Available: https://doi.org/10.1016/j.matpr.2021.07.196

[13] M. A. Sarwar, N. Kamal, W. Hamid, and M. A. Shah, "Prediction of diabetes using machine learning algorithms in healthcare," in *2018 24th International Conference on Automation and Computing*, 2018, pp. 1–6. [Online]. Available: https://doi.org/10.23919/ICONAC.2018.8748992

[14] S. Afolabi, N. Ajadi, A. Jimoh, and I. Adenekan, "Predicting diabetes using supervised machine learning algorithms on e-health records," *Informatics and Health*, vol. 2, no. 1, pp. 9–16, 2025. [Online]. Available: https://doi.org/10.1016/j.infoh.2024.12.002

[15] E. Cengel, E. Cengil, and M. Yildirim, "The role of machine learning models in early diabetes diagnosis: A dataset based analysis," *Computer Science*, vol. 10, no. 1, pp. 33–42, 2025.

[16] M. Maniruzzaman, M. J. Rahman, and M. A.-M. Hasan, "Accurate diabetes risk stratification using machine learning," *Journal of Medical Systems*, vol. 42, no. 5, p. 92, 2018. [Online]. Available: https://doi.org/10.1007/s10916-018-0940-7

[17] A. Mujumdar and V. Vaidehi, "Diabetes prediction using machine learning algorithms," *Procedia Computer Science*, vol. 165, pp. 292–299, 2019. [Online]. Available: https://doi.org/10.1016/j.procs.2020.01.047

[18] M. Maniruzzaman, N. Kumar, and M. M. Abedin, "Comparative approaches for classification of diabetes mellitus data," *Computer Methods and Programs in Biomedicine*, vol. 152, pp. 23–34, 2017. [Online]. Available: https://doi.org/10.1016/j.cmpb.2017.09.004

[19] L. Angeloni and G. Vinti, "Communications on applied nonlinear analysis." [Online]. Available: https://doi.org/10.52783/cana

[20] G. Luo, "Automatically explaining machine learning prediction results," *Health Information Science and Systems*, vol. 4, no. 1, p. 2, 2016. [Online]. Available: https://doi.org/10.1186/s13755-016-0015-4

[21] P. S. Gurunathan, S. Raghuraj, S. Roahit, and S. Nithishkumar, "Web application-based diabetes prediction using machine learning," in *2023 7th International Conference on Computing Methodologies and Communication*, 2023, pp. 296–302. [Online]. Available: https://doi.org/10.1109/ICCMC56507.2023.10083583

[22] S. K. S. Modak and V. K. Jha, "Diabetes prediction model using machine learning techniques," *Multimedia Tools and Applications*, vol. 83, no. 13, pp. 38 523–38 549, 2024. [Online]. Available: https://doi.org/10.1007/s11042-023-16745-4

[23] D. Sisodia and D. S. Sisodia, "Prediction of diabetes using classification algorithms," *Procedia Computer Science*, vol. 132, pp. 1578–1585, 2018. [Online]. Available: https://doi.org/10.1016/j.procs.2018.05.122

[24] G. Tripathi and R. Kumar, "Early prediction of diabetes mellitus using machine learning," in *2020 8th International Conference on Reliability, Infocom Technologies and Optimization*, 2020, pp. 1009–1014. [Online]. Available: https://doi.org/10.1109/ICRITO48877.2020.9197832